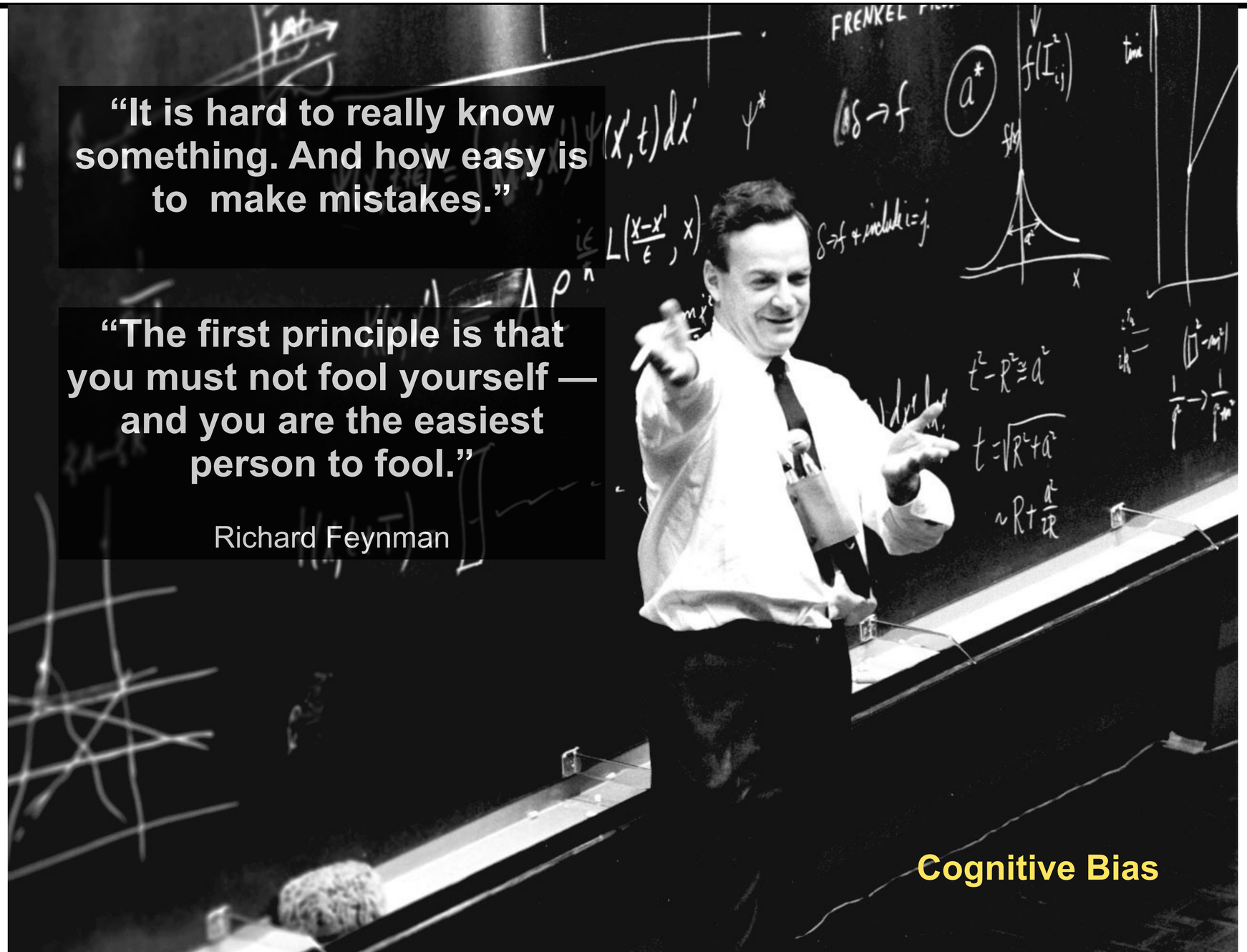
Statistical Inference

Jordi Vitrià
jordi.vitria@ub.edu



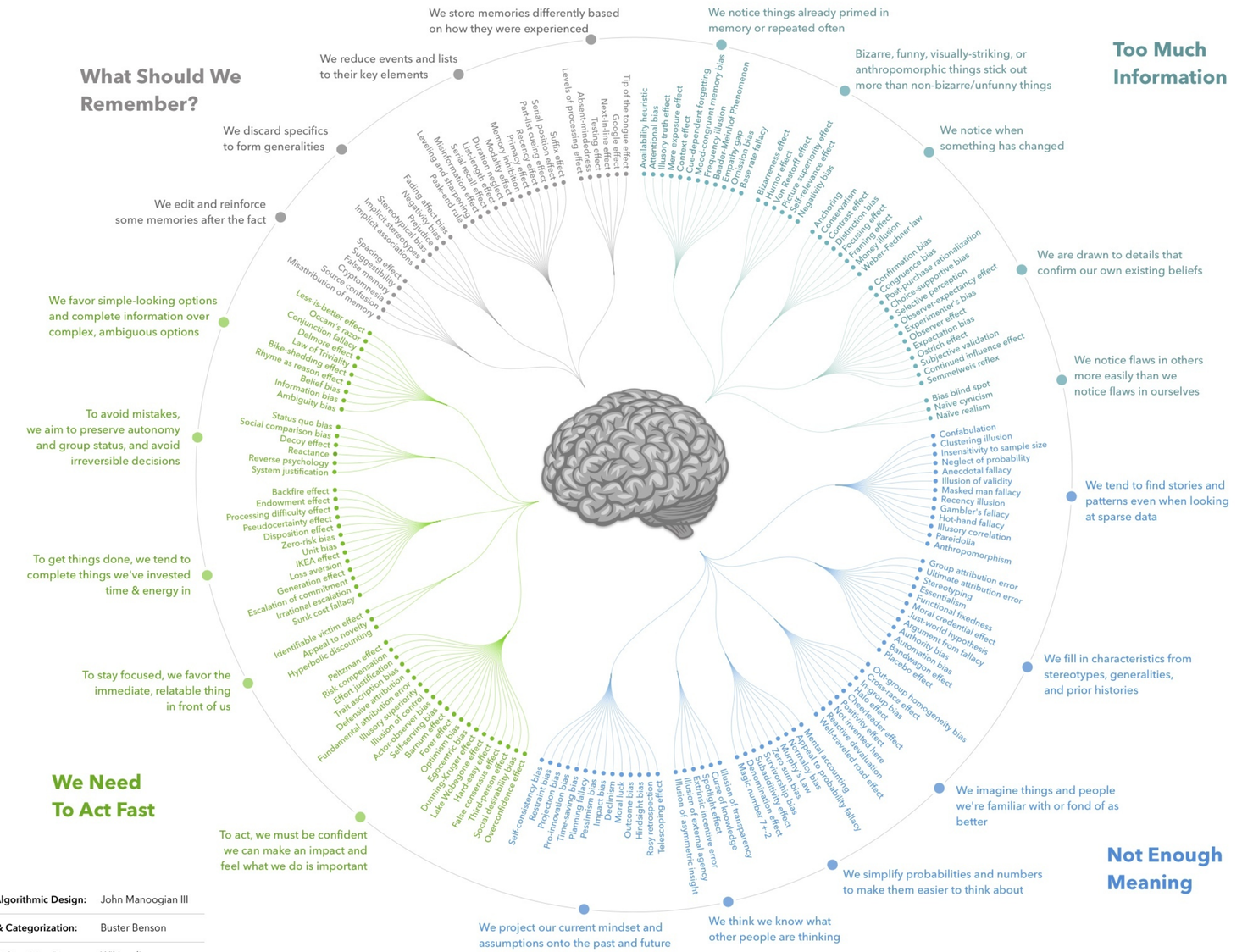
Cognitive Bias

A **cognitive bias** is a systematic pattern of deviation from rationality in judgment.

Rationality = Choosing the best option or decision (in terms of our values, interests and priorities) **based on avalible information about the reality**.

Individuals create their own "subjective reality" from their perception of the world.

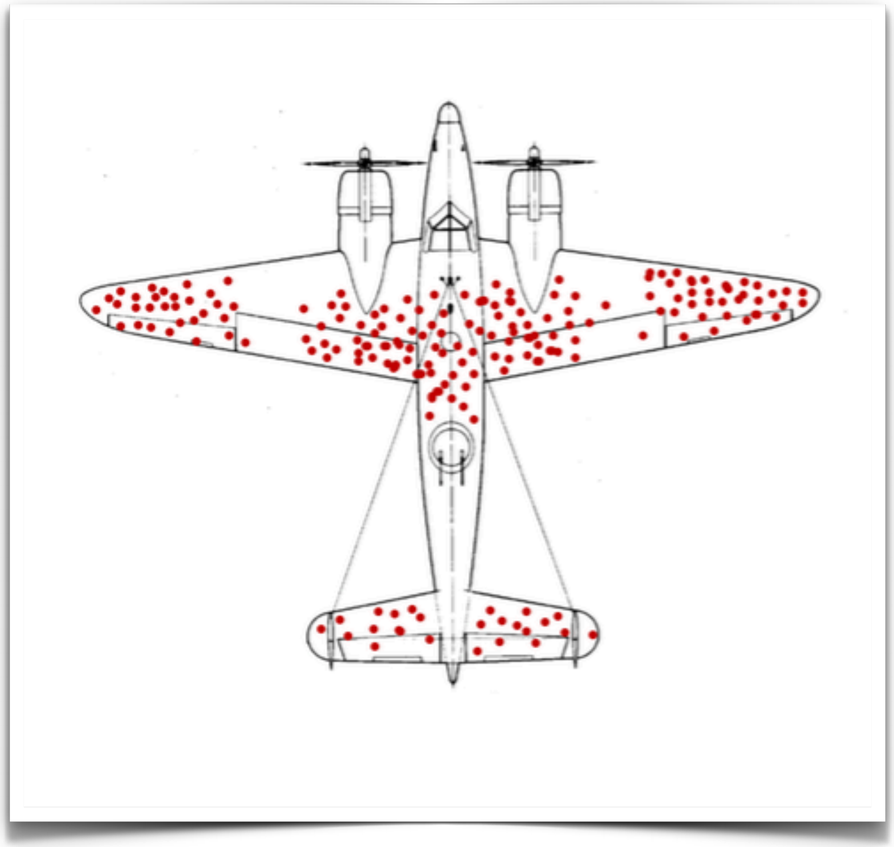
COGNITIVE BIAS CODEX



Visual & Algorithmic Design:	John Manoogian III
Concept & Categorization:	Buster Benson
List of 188 Cognitive Biases:	Wikipedia

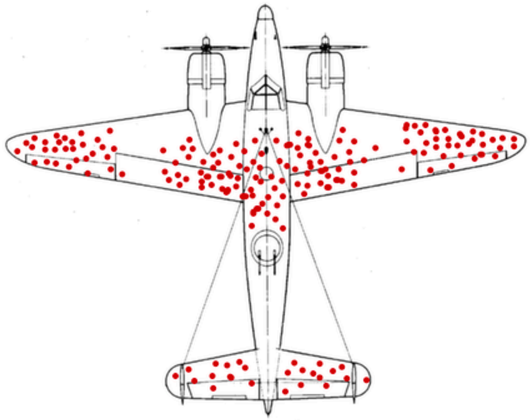
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Survivor Bias

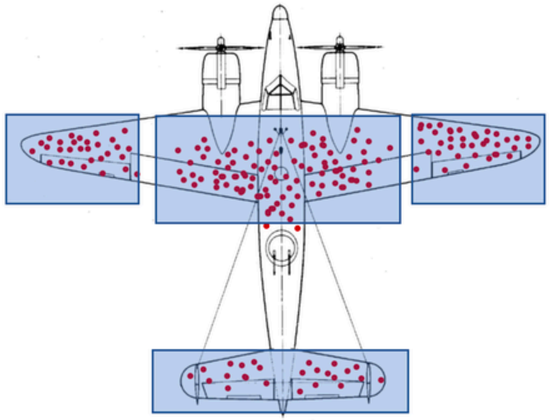


Survivorship bias or **survival bias** is the logical error of concentrating on the people or **things that made it past some selection process and overlooking those that did not**, typically because of their lack of visibility.

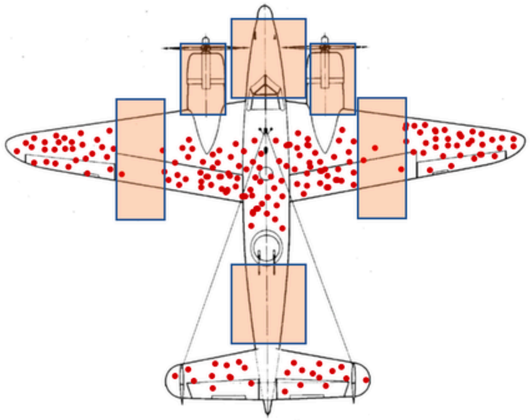
Survivor Bias



Our data if only from returning flights. Here we is a visualization of the places that bullet holes were observed.



And initial guess at how to fix this might be to apply additional armor platting to the parts of the plane with the most holes…



…. However this is where planes that *returned* had bullet holes. The planes we want to protect are the ones that did *not* return, so we should place armor there.

https://www.countbayesie.com/blog/2020/11/5/survivorship-bias-in-house-hunting-a-practical-modeling-example-using-jax



Source:
http://www.nature.com/news/crowdsourced-research-many-hands-make-tight-work-1.18508

Naïve Inductivism Bias

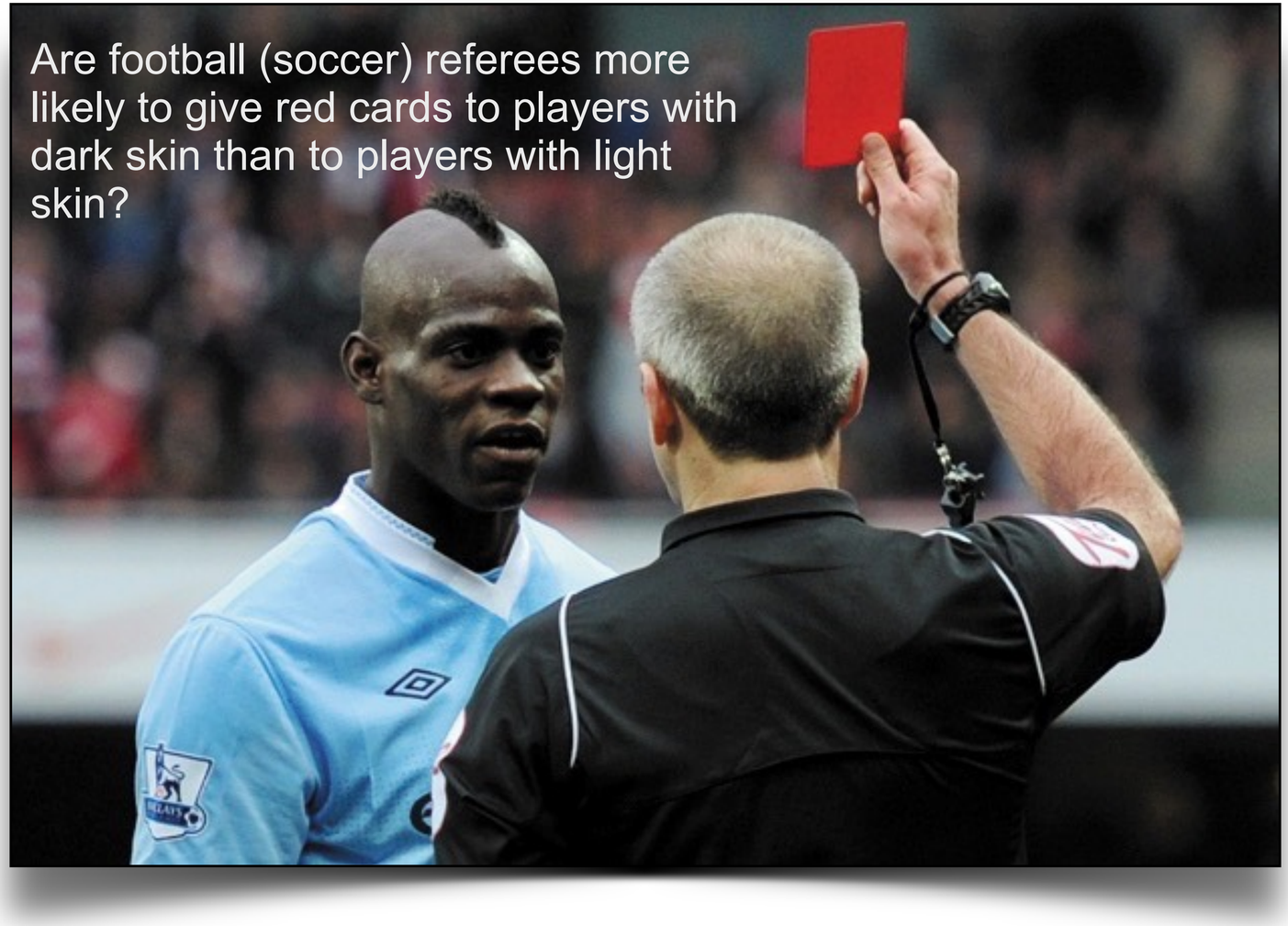
"**Naïve inductivism**": a belief that all scientists seeing the same data should come to the same conclusions.

In "**Of P-Values and Bayes: A Modest Proposal**", Steven N. Goodman, 2001

By implication, anyone who draws a different conclusion must be doing so for nonscientific reasons.

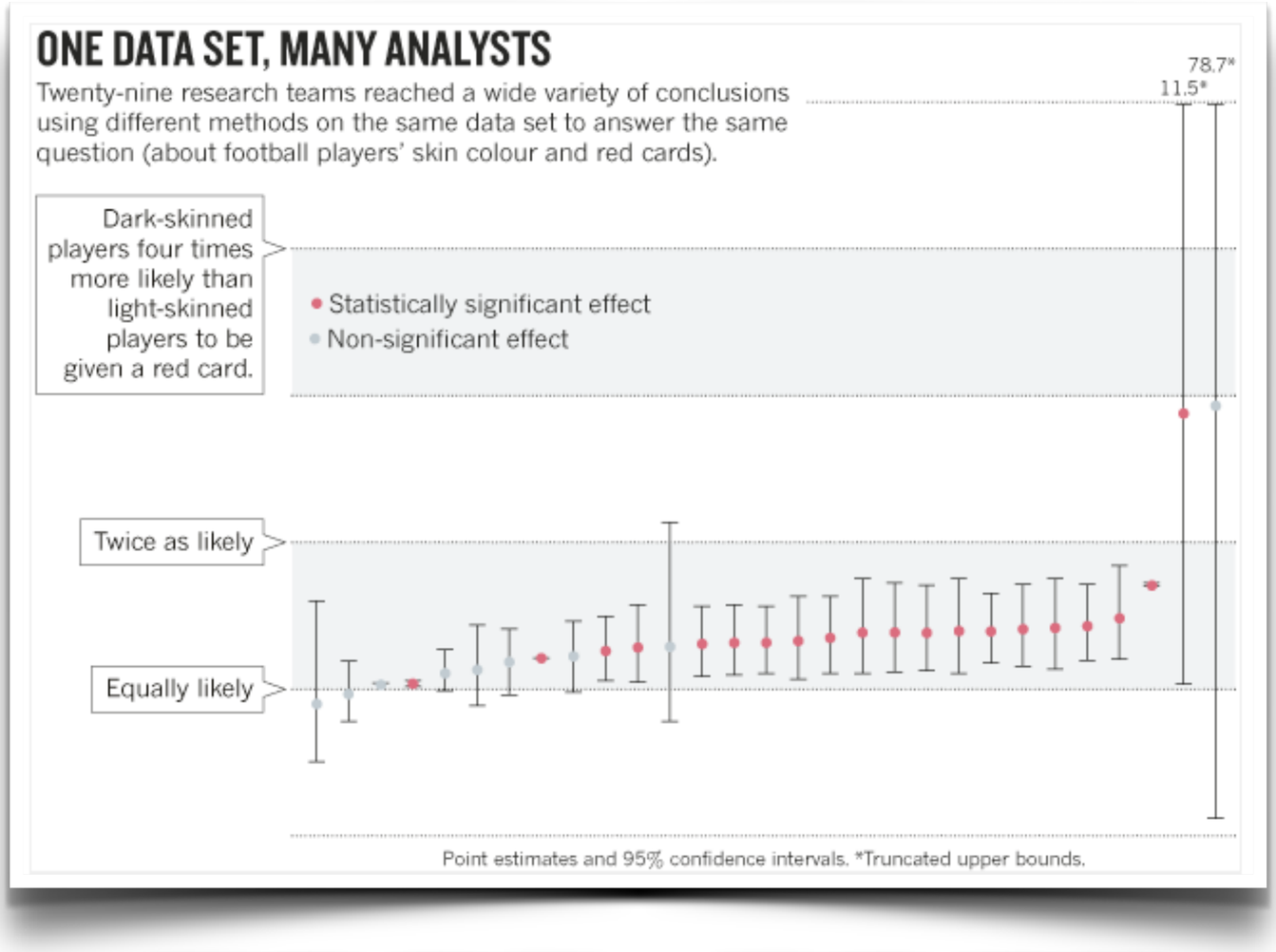
It is a belief that scientific reasoning requires little more than statistical model fitting, or in our case, reporting odds ratios, P-values and the like, to arrive at the truth.

Naïve Inductivism Bias



Source: http://www.nature.com/news/crowdsourced-research-many-hands-make-tight-work-1.18508

Naïve Inductivism Bias



Bad methodology

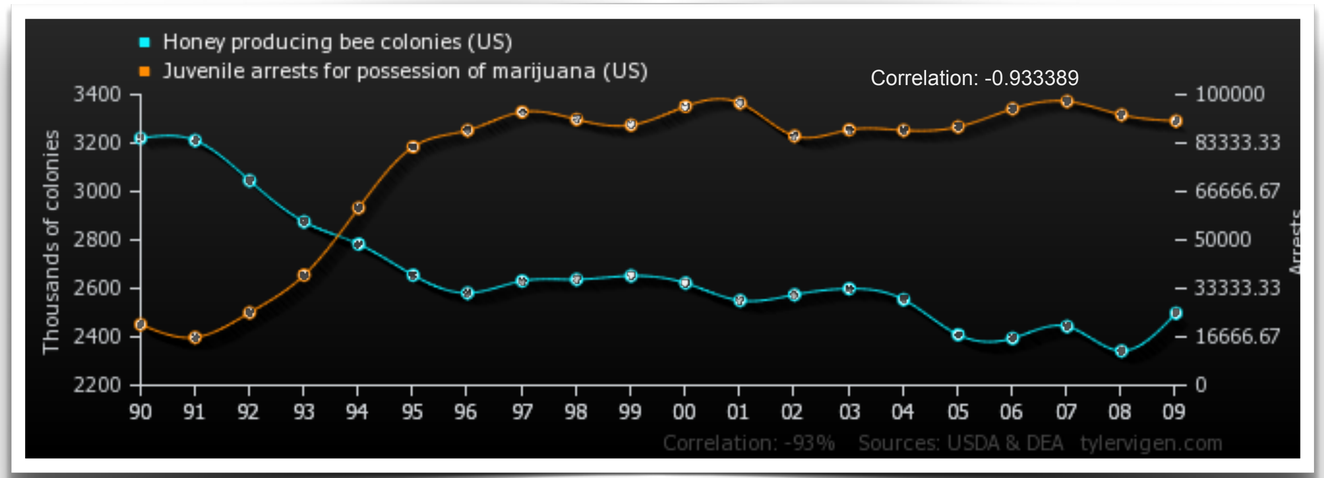
Feature Engineering (Priors)

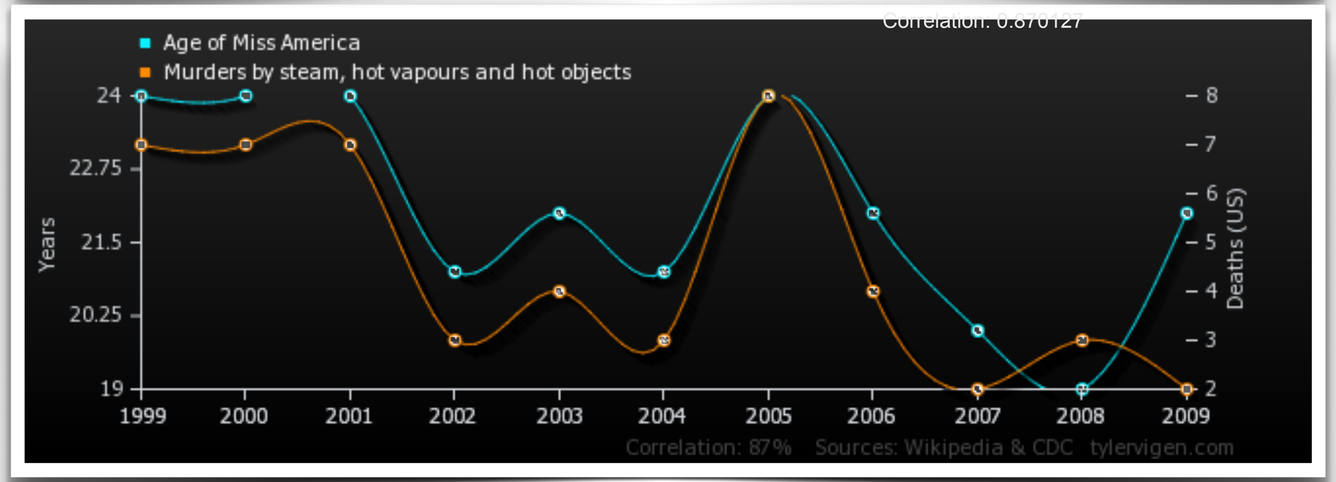


“The numbers have no way of speaking for themselves”

Nate Silver

Apophenia Bias





Source: Spurious Correlations
http://www.tylervigen.com/

Apophenia Bias

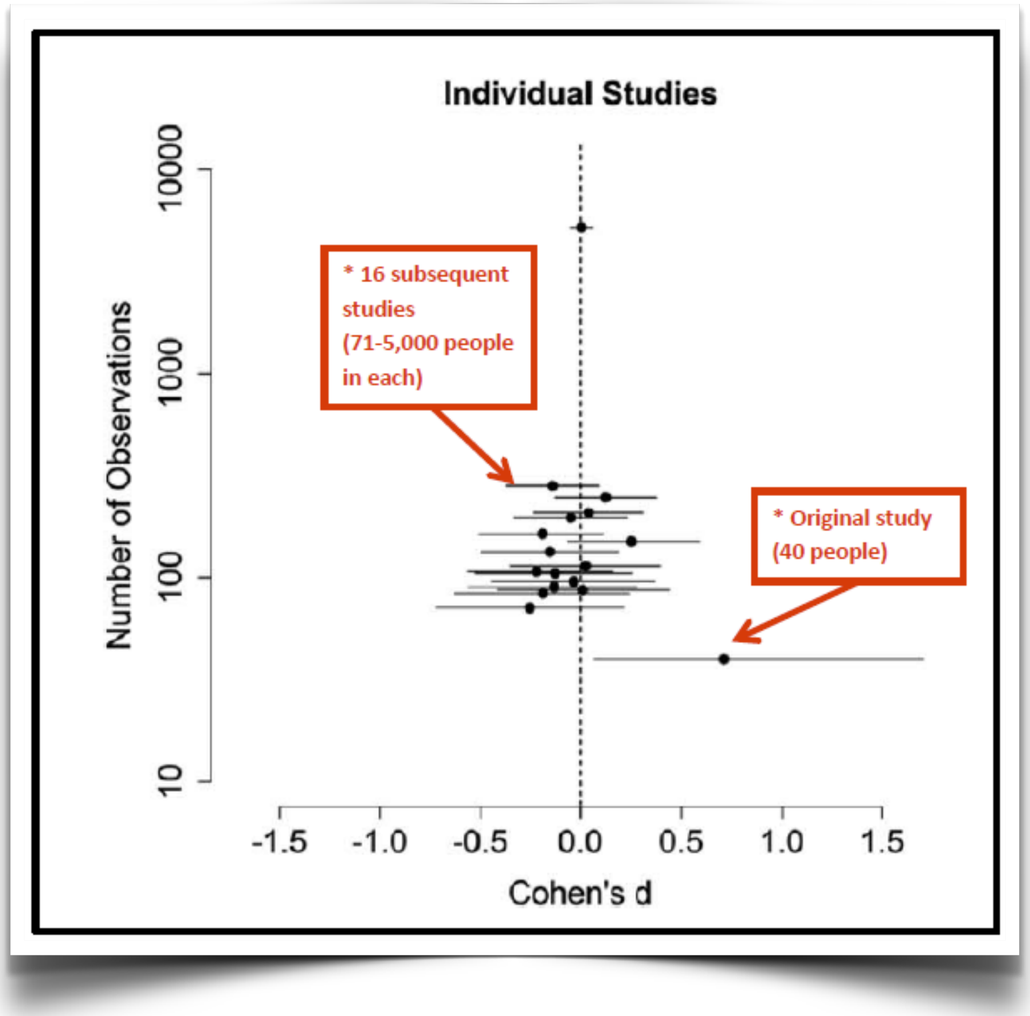
In 2012 Professor Kahneman wrote:

“90% of the students who saw the CRT in normal font made at least one mistake in the test, but the proportion dropped to 35% when the font was barely legible. You read this correctly: performance was better with the bad font.”



http://www.terryburnham.com/2015/04/a-trick-for-higher-sat-scores.html

Apophenia Bias



http://www.terryburnham.com/2015/04/a-trick-for-higher-sat-scores.html

The original paper reached its conclusions based on the test scores of 40 people.

If you analyze a total of over 7,000 people by looking at the original study and 16 additional studies:

Confounding Bias

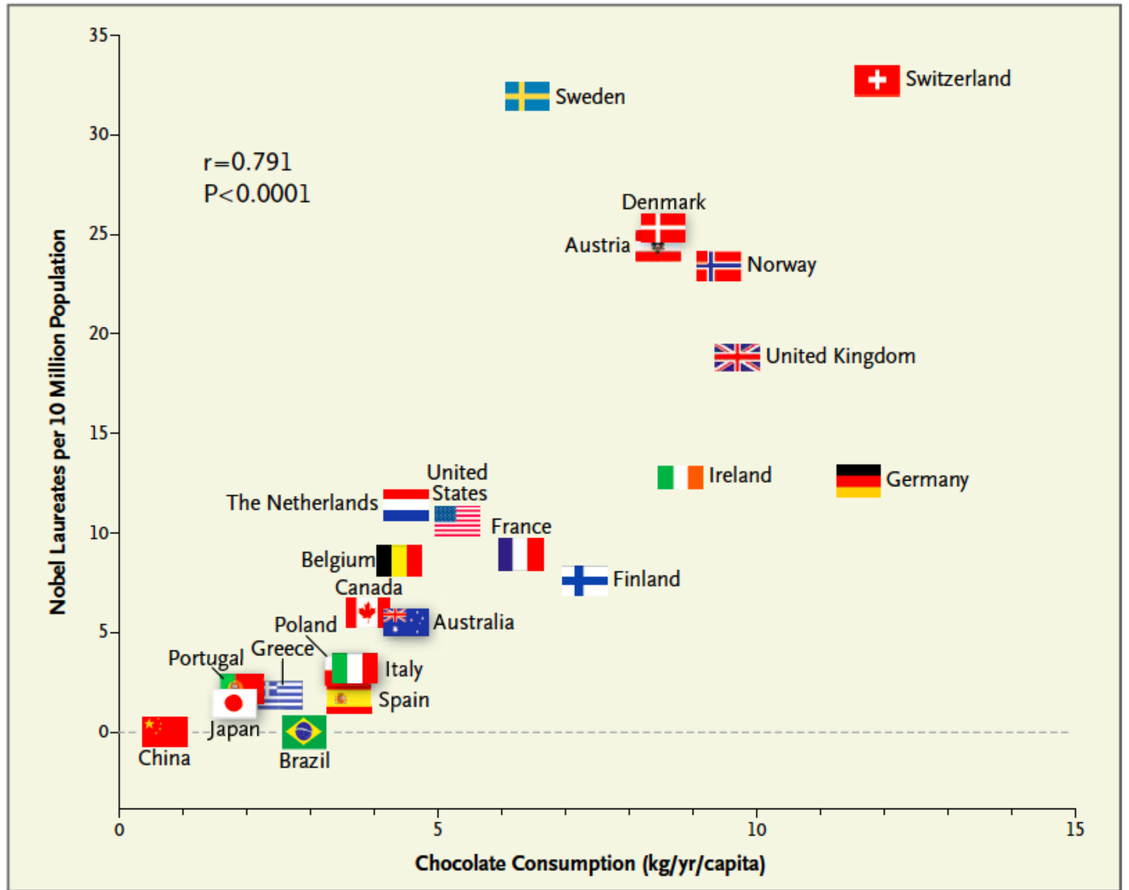


Figure 1. Correlation between Countries' Annual Per Capita Chocolate Consumption and the Number of Nobel Laureates per 10 Million Population.

Confounding Bias

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OCCASIONAL NOTES

Chocolate Consumption, Cognitive Function, and Nobel Laureates

Franz H. Messerli, M.D.

N Engl J Med 2012; 367:1562-1564 | October 18, 2012 | DOI: 10.1056/NEJMon1211064

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Dietary flavonoids, abundant in plant-based foods, have been shown to improve cognitive function. Specifically, a reduction in the risk of dementia, enhanced performance on some cognitive tests, and improved cognitive function in elderly patients with mild impairment have been associated with a regular intake of flavonoids.[1,2] A subclass of flavonoids called flavanols, which are widely present in cocoa, green tea, red wine, and some fruits, seems to be effective in slowing down or even reversing the reductions in cognitive performance that occur with aging. Dietary flavanols have also been shown to improve endothelial function and to lower blood pressure by causing vasodilation in the peripheral vasculature and in the brain.[3,4] Improved cognitive performance with the administration of a cocoa polyphenolic extract has even been reported in aged Wistar–Unilever rats.[5]

Since chocolate consumption could hypothetically improve cognitive function not only in individuals but also in whole populations, I wondered whether there would be a correlation between a country's level of chocolate consumption and its population's cognitive function. To my knowledge, no data on overall national cognitive function are publicly available. Conceivably, however, the total number of Nobel laureates per capita could serve as a surrogate end point reflecting the proportion with superior cognitive function and thereby give us some measure of the overall cognitive function of a given country.

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April 17, 2014 | R.G. Cury and C.H. Moreira

Confounding Bias

Storks Deliver Babies ($p = 0.008$)

KEYWORDS:
Teaching;
Correlation;
Significance;
p-values.

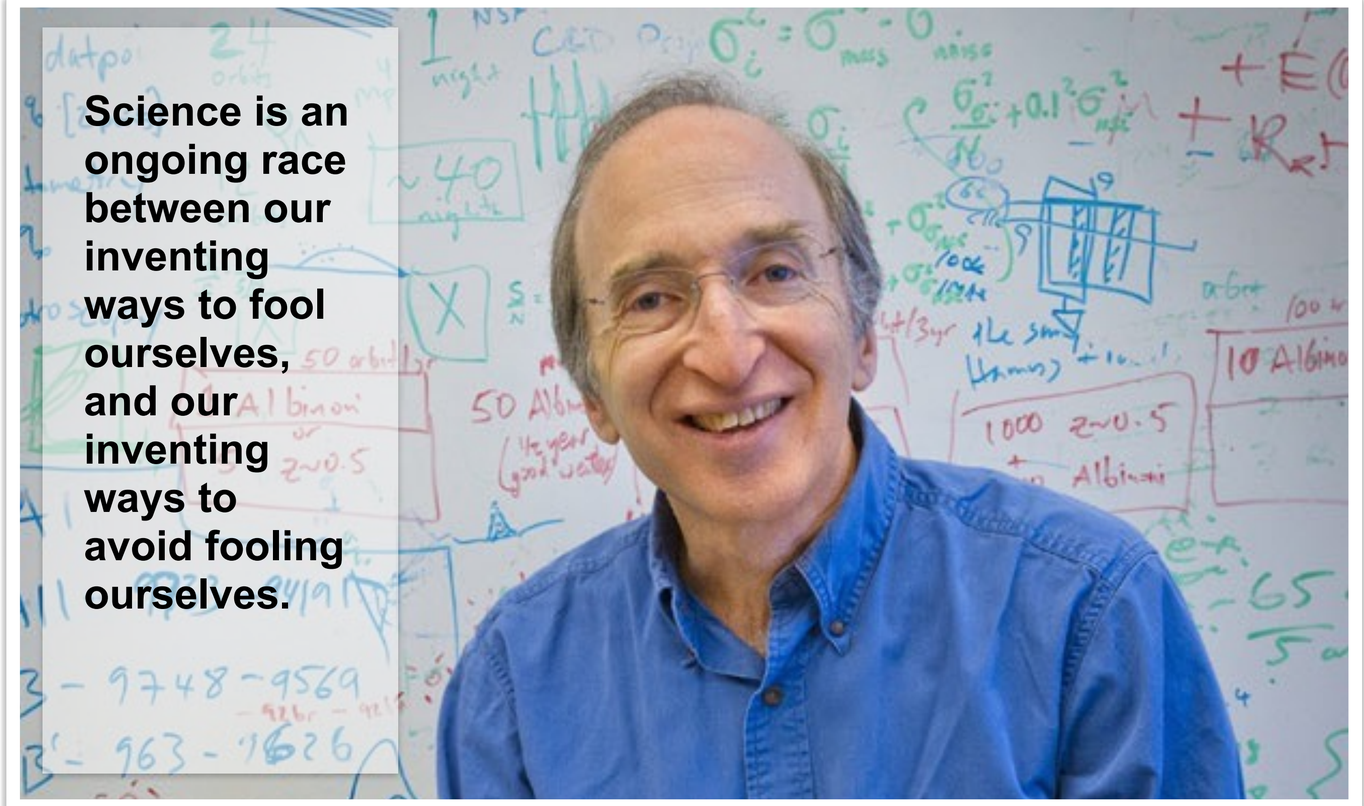
Robert Matthews
Aston University, Birmingham, England.
e-mail: rajm@compuserve.com

Summary
This article shows that a highly statistically significant correlation exists between stork populations and human birth rates across Europe. While storks may not deliver babies, unthinking interpretation of correlation and *p*-values can certainly deliver unreliable conclusions.

Country	Area (km^2)	Storks (pairs)	Humans (10^6)	Birth rate (10^3/yr)
Albania	28,750	100	3.2	83
Austria	83,860	300	7.6	87
Belgium	30,520	1	9.9	118
Bulgaria	111,000	5000	9.0	117
Denmark	43,100	9	5.1	59
France	544,000	140	56	774
Germany	357,000	3300	78	901
Greece	132,000	2500	10	106
Holland	41,900	4	15	188
Hungary	93,000	5000	11	124
Italy	301,280	5	57	551
Poland	312,680	30,000	38	610
Portugal	92,390	1500	10	120
Romania	237,500	5000	23	367
Spain	504,750	8000	39	439
Switzerland	41,290	150	6.7	82
Turkey	779,450	25,000	56	1576

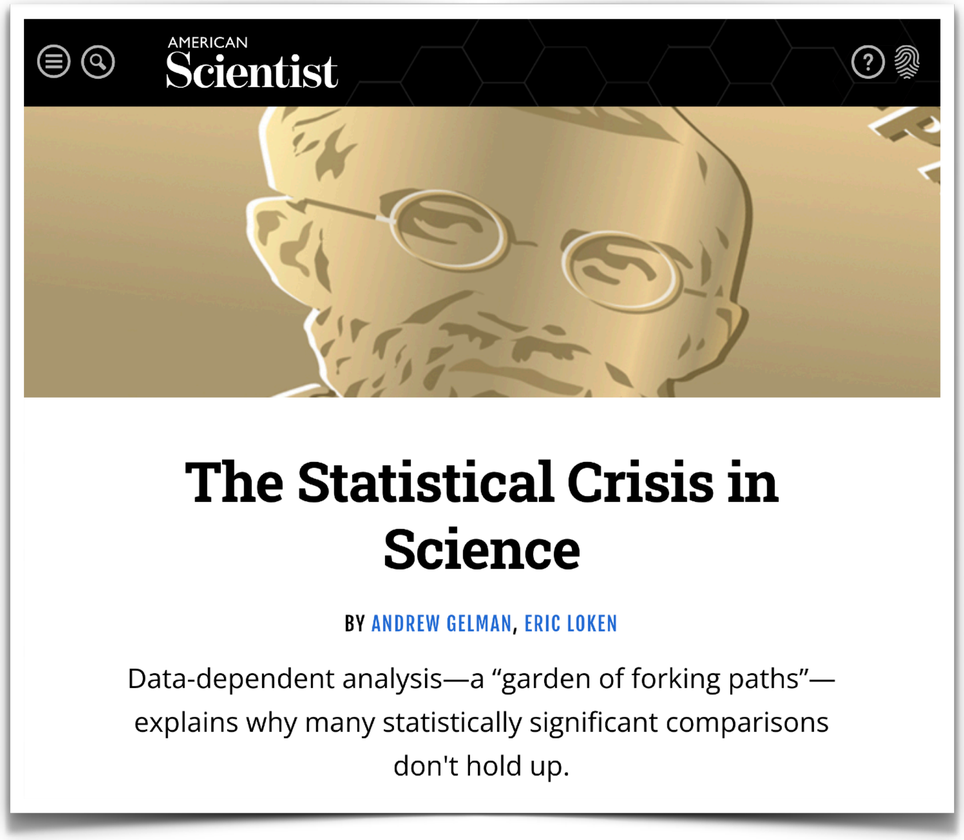
Table 1. Geographic, human and stork data for 17 European countries

Data science is difficult, but not more difficult than general science.



Saul Perlmutter, Astrophysicist, UC Berkeley

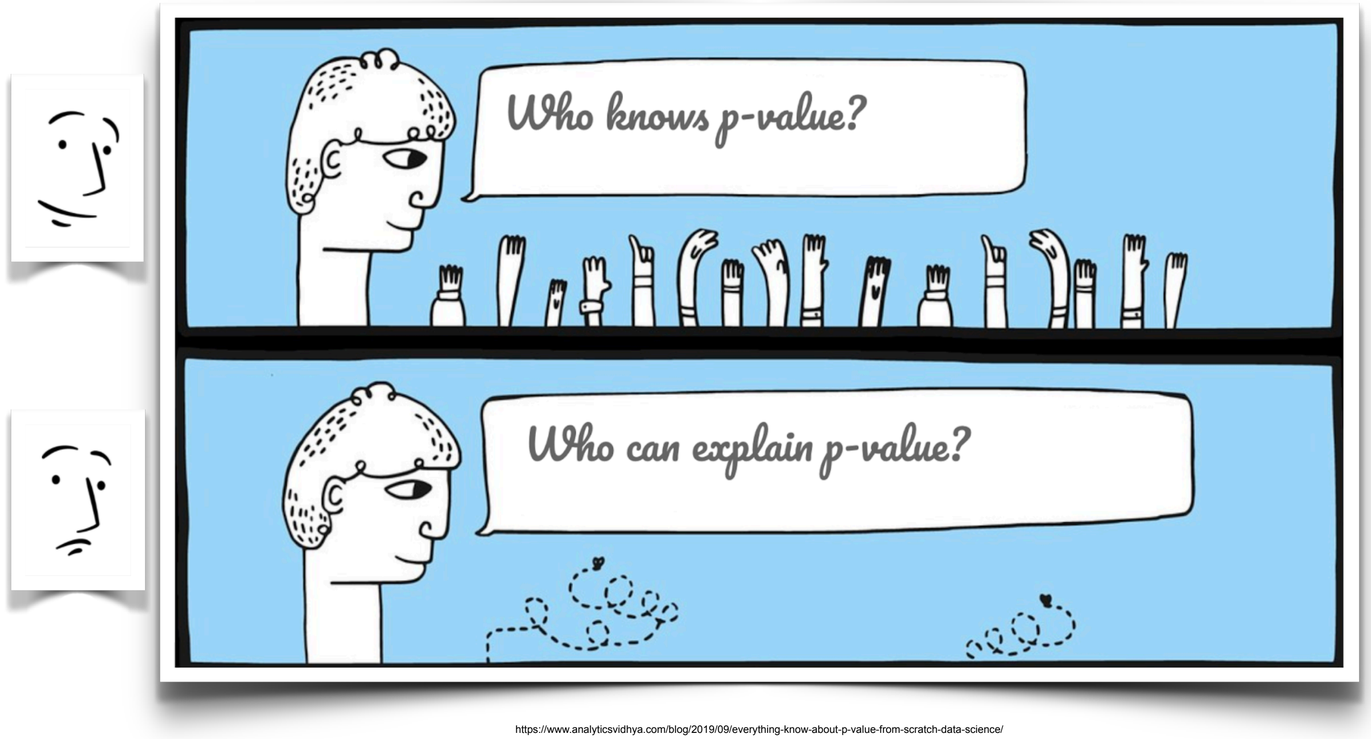
General science has a “reproducibility crisis”





How to check the “validity” of a result?

The “language of probabilities” can be useful for that, but we must be careful.



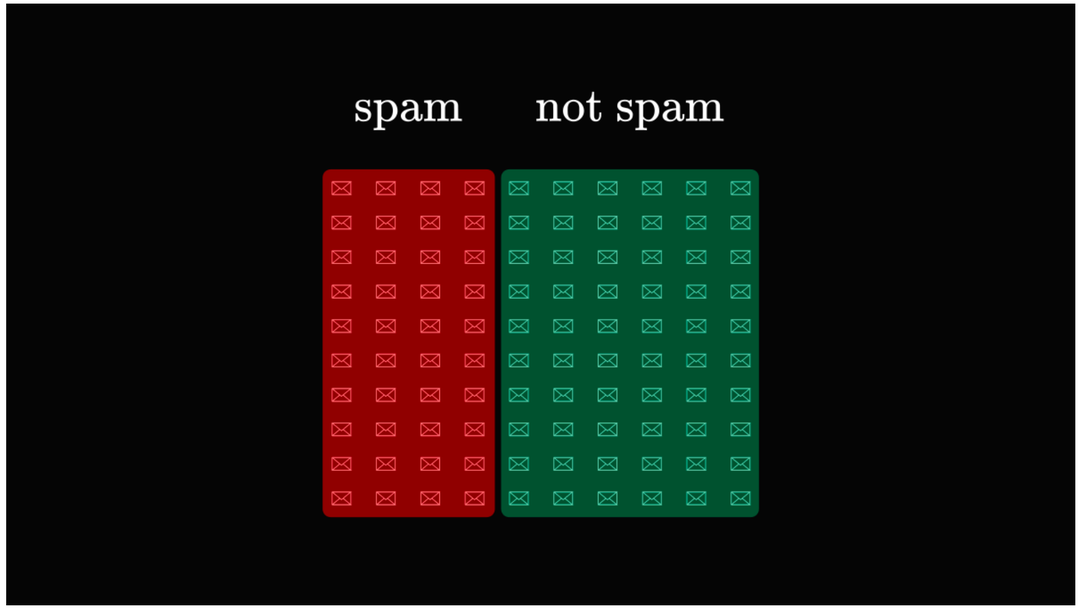
https://www.analyticsvidhya.com/blog/2019/09/everything-know-about-p-value-from-scratch-data-science/

The single most important concept in statistics: Conditional probability

If A and B are two events, they are no necessarily independent of each other. This means that **the occurrence of one can give information about the other**.

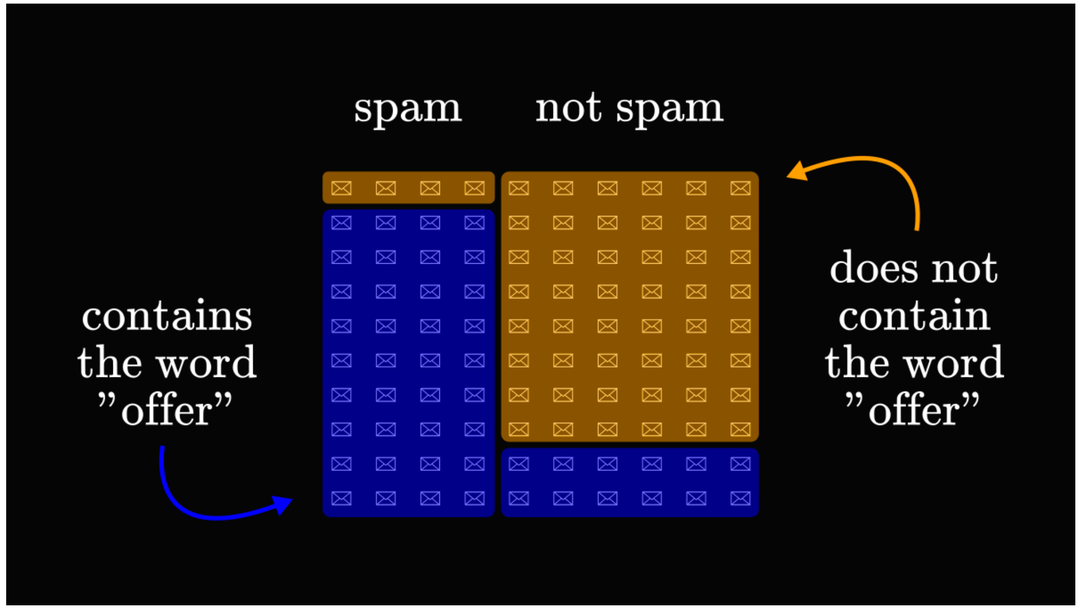
When performing statistical modeling, this is frequently the case.

The single most important concept in statistics: Conditional probability



Suppose that you have 100 emails in your inbox. 40 is spam. 60 is not. Based only on this information, if you receive a random letter, the best guess is a 40% chance of it being spam.

The single most important concept in statistics: Conditional probability



What about taking a look at the mail and checking for the word "offer"? It turns out that among the 48 mails containing this word, 36 are spam!

The single most important concept in statistics:
Conditional probability

Let's formulate this in the language of probability! Based on this sample, if you know that a mail contains the word "offer", there is a 75% chance of being spam. In mathematical words, we have:

$$P(\textbf{spam} \mid \textbf{"offer"}) = \frac{36}{48} = 0.75$$

Essentially, conditional probability restricts the event space to the observed event. It allows you to refine your statistical models upon gaining new information.

The single most important concept in statistics:
Conditional probability

You have somebody let's say he is a senator, US senator…

He's very moderate, he's intelligent, rectitude is the word that comes to mind for everybody that thinks of him. He lives modestly with his wife and his daughter who's unfortunately very sick.

Given that background, what's more likely that this senator took a bribe from a lobbyist, or that he took a bribe from a lobbyist and used the money to pay for his daughter's expensive operation?

WHAT IS STATITICAL INFERENCE?

Descriptive statistics describes a sample, **inferential statistics** infer predictions about a population.

Statistical inference deals with parameter estimation and understanding the quality of this estimation.

Frequentist paradigm assumptions vs Bayesian paradigm assumptions

There are two main approaches to statistical inference: **Frequentist and Bayesian.**

Frequentist assumptions:

- Data are a repeatable random sample - there is a frequency.
- Underlying assumptions remain constant during this repeatable process;
- Population parameters are fixed.

Bayesian assumptions:

- Data is observed from a realized sample.
- Data are fixed.
- Parameters are unknown and described probabilistically.

Frequentist paradigm assumptions vs Bayesian paradigm assumptions

P-values and confidence intervals are the most common propositions of frequentist statistics.

Statistical inference is based on assumptions related to a statistical model or sampling process. Correct inference requires these assumptions to be correct.

Today we will develop the **frequentist inference paradigm**. It bases the production of propositions by considering **repeated sampling of a population**.